

Calculus 2, Moed A 2025

Reichman university Solutions

Lecturer: Dr. Yossi Shamaï. Duration: 3 hours.

Auxiliary material: none.

Instructions:

The following exam has 5 questions. You must answer exactly 4 questions. Each question is worth 25 credits. There is no choice between different items of the same question.

Unexplained answers will not receive any credit. You must state all of the conditions for every theorem and/or claim. You do not have a choice between different items of the same question. It is forbidden to rely on any material that was not learned in the course. You must write your solutions only in the designated areas. Do not write at the back of the papers!

Good Luck!

1. (25 points)

1.1. (17 points) Let $a, b \in \mathbb{R}$ such that $a < b$ and let f, G be two functions that are defined on $[a, b]$.

1.1.1. (2 points) Define the following term: The function G is an anti-derivative of the function f in $[a, b]$.

Solution:

We say that G is an anti-derivative of f in $[a, b]$ if G is differentiable on $[a, b]$ and if for every $x \in [a, b]$ we have

$$G'(x) = f(x).$$

1.1.2. (15 points) Suppose that f is continuous on $[a, b]$ and that G is an anti-derivative of f in $[a, b]$. Prove that

$$\int_a^b f(x) dx = G(b) - G(a)$$

Solution:

See lectures.

1.2. (8 points) Compute the value of the following series, or prove that it does not converge in the extended sense:

$$\sum_{n=1}^{\infty} \left[\arcsin\left(\frac{1}{\sqrt{n+1}}\right) - \arcsin\left(\frac{1}{\sqrt{n}}\right) \right]$$

Solution:

Note that

$$\begin{aligned} \sum_{n=1}^{\infty} \left[\arcsin\left(\frac{1}{\sqrt{n+1}}\right) - \arcsin\left(\frac{1}{\sqrt{n}}\right) \right] &= \lim_{N \rightarrow \infty} \sum_{n=1}^N \left[\arcsin\left(\frac{1}{\sqrt{n+1}}\right) - \arcsin\left(\frac{1}{\sqrt{n}}\right) \right] \\ &= \lim_{N \rightarrow \infty} \left[\cancel{\arcsin\left(\frac{1}{\sqrt{2}}\right)} - \arcsin(1) + \cancel{\arcsin\left(\frac{1}{\sqrt{3}}\right)} - \cancel{\arcsin\left(\frac{1}{\sqrt{2}}\right)} + \cdots + \arcsin\left(\frac{1}{\sqrt{N+1}}\right) - \cancel{\arcsin\left(\frac{1}{\sqrt{N}}\right)} \right] \\ &= \lim_{N \rightarrow \infty} \left[\arcsin\left(\frac{1}{\sqrt{N+1}}\right) - \arcsin(1) \right] = -\arcsin(1) = -\frac{\pi}{2} \end{aligned}$$

2. (25 points)

2.1. (15 points) Prove that the number e is irrational.

Solution: See lectures.

2.2. (10 points) Determine whether the following improper integral converges or diverges:

$$\int_1^{\infty} e^{-x} \ln(x) dx$$

Solution:

First, we clearly have $e^{-x} \ln(x) \geq 0$ for every $x \geq 1$. Thus we can use the comparison test. Note that

$$\lim_{x \rightarrow \infty} x e^{-x} = \lim_{x \rightarrow \infty} \frac{x}{e^x} \stackrel{\frac{\infty}{\infty}}{=} \lim_{x \rightarrow \infty} \frac{1}{e^x} = 0 \quad (\star)$$

We know that $\ln(1+x) \leq x$ for every $x > -1$. Thus, $\ln(x) \leq x-1 \leq x$ for every $x \geq 1$. We now compute

$$\int_1^{\infty} e^{-x} \ln(x) dx \leq \int_1^{\infty} x e^{-x} dx \stackrel{\text{IBP}}{=} [-x e^{-x}]_1^{\infty} + \int_1^{\infty} e^{-x} dx \stackrel{(\star)}{=} \frac{1}{e} + [-e^{-x}]_1^{\infty} = \frac{2}{e}$$

Thus, the integral converges by the comparison test.

3. (25 points)

3.1. (13 points) Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} \frac{\arctan x}{x} & x \neq 0 \\ 1 & x = 0 \end{cases} \quad \forall x \in \mathbb{R}$$

3.1.1. (9 points) Find the Taylor series about $x_0 = 0$ of f and find its domain of convergence.

Solution:

We know that $\arctan(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$ for every $-1 \leq x \leq 1$. Dividing both sides by x , then for every $-1 \leq x \leq 1$ such that $x \neq 0$ we get

$$\frac{\arctan(x)}{x} = 1 - \frac{x^2}{3} + \frac{x^4}{5} - \dots$$

As $f(0) = 1$, it follows that for every $-1 \leq x \leq 1$ we have

$$f(x) = 1 - \frac{x^2}{3} + \frac{x^4}{5} - \dots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{2n+1}$$

By the theorem of uniqueness of power series, we conclude that the Taylor series of f is $T_f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{2n+1}$ and that the range of convergence is $[-1, 1]$.

3.1.2. (4 points) Compute the value of $f^{(10)}(0)$.

Solution:

As $f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{2n+1}$ for every $-1 \leq x \leq 1$, then by the theorem of uniqueness, for every $0 \leq n \in \mathbb{N}$ it holds that $a_n = \frac{f^{(n)}(0)}{n!}$, and therefore $f^{(n)}(0) = n! \cdot a_n$, where a_n is the coefficient of x^n . It follows that

$$f^{(10)}(0) = -\frac{10!}{11}$$

3.2. (12 points) Let $(a_n)_{n=0}^{\infty}$ be a sequence of real numbers, and let R be the convergence radius of the power series $\sum_{n=0}^{\infty} a_n x^n$. Suppose that $R < 1$. Prove or disprove the following statement: It holds that $\lim_{n \rightarrow \infty} a_n \neq 0$.

Solution:

True. ATC that $\lim_{n \rightarrow \infty} a_n = 0$. Thus, there exists an $N_1 \in \mathbb{N}$ such that for every $n \geq N_1$ we have

$$|a_n - 0| < \boxed{1} \Rightarrow |a_n| < 1$$

We know that $\sum_{n=N_1}^{\infty} x^n$ converges absolutely iff $-1 < x < 1$.

Let $-1 < x_1 < 1$. We note that

$$\sum_{n=N_1}^{\infty} |a_n x_1^n| \leq \sum_{n=N_1}^{\infty} |x_1^n|$$

Thus the series $\sum_{n=N_1}^{\infty} a_n x^n$ converges absolutely at x_1 .

Thus the series $\sum_{n=N_1}^{\infty} a_n x^n$ converges absolutely for every $-1 < x < 1$ and therefore the series $\sum_{n=0}^{\infty} a_n x^n$ converges absolutely for every $-1 < x < 1$, which implies that $R \geq 1$. Contradiction!

4. (25 points)

4.1. (13 points) Determine whether the following series is convergent or divergent:

$$1 - \frac{1}{4} - \frac{1}{3} + \frac{1}{3} - \frac{1}{8} - \frac{1}{6} + \frac{1}{5} - \frac{1}{12} - \frac{1}{9} + \cdots + \frac{1}{2n-1} - \frac{1}{4n} - \frac{1}{3n} + \cdots$$

Solution:

We prove that the series is divergent. ATC that the series is convergent. Then $\exists s \in \mathbb{R}$ such that

$$s = 1 - \frac{1}{4} - \frac{1}{3} + \frac{1}{3} - \frac{1}{8} - \frac{1}{6} + \frac{1}{5} - \frac{1}{12} - \frac{1}{9} + \cdots + \frac{1}{2n-1} - \frac{1}{4n} - \frac{1}{3n} + \cdots$$

Consider the series obtained from the series above where terms are grouped by adding parentheses on every three terms.

Then by the parentheses test

$$\begin{aligned} s &= \left(1 - \frac{1}{4} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{8} - \frac{1}{6}\right) + \cdots + \left(\frac{1}{2n-1} - \frac{1}{4n} - \frac{1}{3n}\right) + \cdots \\ &= \sum_{n=1}^{\infty} \left(\frac{1}{2n-1} - \frac{1}{4n} - \frac{1}{3n}\right) = \sum_{n=1}^{\infty} \left(\frac{1}{2n-1} - \frac{7}{12n}\right) \\ &= \sum_{n=1}^{\infty} \frac{7-2n}{24n^2-12n} = -\sum_{n=1}^{\infty} \frac{2n-7}{24n^2-12n} \end{aligned}$$

We consider the series $\sum_{n=4}^{\infty} \frac{2n-7}{24n^2-12n}$, which is nonnegative. Note that by comparison with $b_n = \frac{1}{n}$, we have

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{n(2n-7)}{24n^2-12n} = \frac{1}{12}$$

which implies that the series $\sum_{n=4}^{\infty} \frac{2n-7}{24n^2-12n}$ is divergent along with the harmonic series. This implies that the series $\sum_{n=1}^{\infty} \frac{2n-7}{24n^2-12n}$ is divergent (has a divergent tail), and by the algebra of converging series, the series $\sum_{n=1}^{\infty} \frac{7-2n}{24n^2-12n}$ is divergent.

- 4.2. (12 points) Let f be a function that is continuously differentiable on $[0, \pi]$. Suppose that $\int_0^{\pi} x f'(x) dx = \int_0^{\pi} f(x) dx$. Suppose also that $f(\pi) = 1$. Compute the value of the definite integral $\int_0^{\pi} f(x) dx$.

Hint: first explain why the integrals exist.

Solution:

First, as f is continuously differentiable on $[0, \pi]$ then $x f'(x)$ is continuous on $[0, \pi]$ and therefore integrable on $[0, \pi]$. Thus, the integral $\int_0^{\pi} x f'(x) dx$ exists. Also, as f is differentiable on $[0, \pi]$ then f is continuous on $[0, \pi]$ and therefore integrable on $[0, \pi]$. Thus, the integral $\int_0^{\pi} f(x) dx$ exists. Note that

$$\int_0^{\pi} f(x) dx = \int_0^{\pi} x f'(x) dx \stackrel{\text{IBP}}{=} [x \cdot f(x)]_0^{\pi} - \int_0^{\pi} f(x) dx = \pi f(\pi) - \int_0^{\pi} f(x) dx$$

It follows that $\int_0^{\pi} f(x) dx = \frac{\pi \cdot f(\pi)}{2} = \frac{\pi}{2}$.

5. (25 points)

- 5.1. (13 points) Let $f : [1, \infty) \rightarrow \mathbb{R}$ be a twice differentiable function. Suppose that:

- f is concave (namely, $f''(x) \leq 0$ for every $x \geq 1$).
- $\lim_{x \rightarrow \infty} f'(x) = 0$.
- The series $\sum_{n=1}^{\infty} f'(n)$ is convergent.

Prove that $\lim_{x \rightarrow \infty} f(x)$ exists.

Solution:

First, as f is concave, then $f''(x) \leq 0$ for every $x \geq 0$ and therefore f' is decreasing. As $\lim_{x \rightarrow \infty} f'(x) = 0$ then the conditions for the integral test hold, and as the series $\sum_{n=1}^{\infty} f'(n)$ is convergent, then the integral $\int_1^{\infty} f'(x) dx$ is convergent. Thus,

$$\lim_{x \rightarrow \infty} f(x) = f(1) + \lim_{x \rightarrow \infty} [f(x) - f(1)] = f(1) + \int_1^{\infty} f'(x) dx$$

- 5.2. (12 points) Determine whether the following series converges or diverges:

$$\sum_{n=1}^{\infty} \frac{(n!)^2}{2 \cdot 4 \cdot 6 \cdots (4n)}$$

Hint: Ensure that you understand the pattern of the series' terms before writing your solution.

Solution:

We use the ratio test,

$$L = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{[(n+1)!]^2 \cdot 2 \cdot 4 \cdot 6 \cdots (4n)}{2 \cdot 4 \cdot 6 \cdots (4n) \cdot (4n+2)(4n+4) \cdot (n!)^2} = \lim_{n \rightarrow \infty} \frac{(n+1)^2}{(4n+2)(4n+4)} = \lim_{n \rightarrow \infty} \frac{\left(1 + \frac{1}{n}\right)^2}{\left(4 + \frac{2}{n}\right)\left(4 + \frac{4}{n}\right)} = \frac{1}{16}$$

It follows that $L < 1$ and consequently, the series is convergent.

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